Compiled Problems - Intermediate Division

1. **01-02 C1 Recursive Functions**

   Find f(7)

$$f(x) = \begin{cases} f(x-2)+2 & \text{if } x > 2x+2 \\ \text{otherwise} \end{cases}$$

2. **01-02 C1 Recursive Functions**

   Find f(7,4)

$$f(x) = \begin{cases} x - y & \text{if } x \le 0 \\ x - f(x-2, y-1) & \text{if } x > 0 \text{ and } x \text{ is even} \\ y - f(y-2, x-1) & \text{if } x > 0 \text{ and } x \text{ is odd} \end{cases}$$

3. **02-03 C1 Recursive Functions**

   Find F(8)

$$F(x) = \begin{cases} F(F(x-3)) + 4 & \text{if } x > 5 \\ 2x - 3 & \text{if } x \le 5 \end{cases}$$

4. **02-03 C1 Recursive Functions**

   Find F(6) expressed in a reduced fractional form.

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } x \le 1 \\ x / (x+1) * F(x-1) + 1 & \text{if } x > 1 \end{cases}$$

5. **03-04 C1 Recursive Functions**

   Find f(11,7)

$$f(x,y) = \begin{cases} f(y - x, x - 1) + 3 & \text{if } x > y \\ x - y & \text{otherwise} \end{cases}$$

6. **03-04 C1 Recursive Functions**

Find f (8)

$$f(x) = \begin{cases} f(f(x-3)) + 4 & \text{if } x > 4 \\ x^2 - 3 & \text{if } x \le 4 \end{cases}$$

7. **04-05 C1 Recursive Functions**

Find f(12)

$$f(x) = \begin{cases} f(x-2) + x & \text{if } x \text{ is a multiple of 3} \\ (x-3) - x & \text{if } x \text{ is a multiple of 5} \\ 2x + 5 & \text{otherwise} \end{cases}$$

8. **04-05 C1 Recursive Functions**

Find f(f(f(7)))

$$f(x) = \begin{cases} f(x-2) + 1 & \text{if } x \ge 5 \\ f(x-1) - x & \text{if } 2 < x < 5 \\ 2x & \text{if } x \le 2 \end{cases}$$

9. **05-06 C1 Recursive Functions**

Find F(8,12)

$$F(X, Y) = \begin{cases} F(X+1, Y-2) + 3 & \text{if } X<Y \\ X^2 + Y^2 & \text{if } X=Y \\ F(Y-1, X) - 1 & \text{if } X>Y \end{cases}$$

10. **05-06 C1 Recursive Functions**

Start with a segment 16 cm in length. At each end draw a segment half as long, perpendicular to the original segment and intersecting at the new segment's midpoint. If this process is continued infinitely, a fractal is created. Counting the original segment as stage 1, find the length of each segment added during stage 7.

11. **06-07 C1 Recursive Functions**

Find $f(f(f(f(f(10)))))$

$$f(x)=\begin{cases} f(x-5)-3 & \text{if } x>7 \\ f(x+2)+2 & \text{if } 3<x\le 7 \\ x+5 & \text{if } x\le 3 \end{cases}$$

12. **06-07 C1 Recursive Functions**

Find $f(30,12)$ where max is the bigger result of the 2 arguments.

$$f(x,y)=\begin{cases} \max\big(f(x-4,y+3),f(y,x)\big) & \text{if } x>y \\ x*y & \text{otherwise} \end{cases}$$

13. **07-08 C1 Recursive Functions**

Find $f(32)$ given:

$$f(x)=\begin{cases} f(x/2-1)+3 & \text{if } x \text{ is even} \\ 2*f(x-3)-5 & \text{if } x \text{ is odd and positive} \\ x^2-3 & \text{if } x \text{ is odd and negative} \end{cases}$$

14. **07-08 C1 Recursive Functions**

If $f(1) = 8$ and $f(n+1) = 2*f(n) - 4$, find $f(5)$.

15. **08-09 C1 Recursive Functions**

Find $f(35,8)$ given:

$$f(x,y)=\begin{cases} f(x-y,y+2)+y & \text{if } x\ge y \\ x^2-y & \text{if } x<y \end{cases}$$

16. **08-09 C1 Recursive Functions**

Find the sum of $f(1)+f(2)+f(3)+\ldots+f(10)$ given:
$f(1) = 1$ and $f(n) = f(n-1) + 5$

17. **09-10 C1 Recursive Functions**

Find $f(13)$ given:

$$f(x)=\begin{cases} f(x-1)+f(x-2)+2 & \text{if } 7\le x\le 10 \\ 2*f(x-3)-7 & \text{if } x>10 \\ x-5 & \text{if } x<7 \end{cases}$$

18. **09-10 C1 Recursive Functions**

Find $f(30)$ given:

$$f(x)=\begin{cases}3 & \text{if } x=1\\ 7 & \text{if } x=2\\ f(x-1)+4 & \text{if } x\geq 3\end{cases}$$

19. 1**0-11 C1 Recursive Functions**

Find $f(18)$ given:

$$f(x)=\begin{cases}f(x-5)+3 & \text{if } x\geq 10\\ \text{x}+6 & \text{o}\textit{therwise}\\ f(2x-1)-2 & \text{if } 0\leq x\leq 8\end{cases}$$

20. 1**0-11 C1 Recursive Functions**

Find $f(f(f(f(18))))$ given:

$$f(x)=\begin{cases}f(x-5)-2 & \text{if } x\geq 10\\ f(x+4)+6 & \text{if } 7\leq x<10\\ x-4 & \text{if } x<7\end{cases}$$

21. 1**1-12 C1 Recursive Functions**

Find $f(72)$ given:

$$f(x)=\begin{cases}f([x/2])+x & \textit{if } x\geq 10\\ 2x & \textit{if } x<10\end{cases}$$

where [n] represents the greatest integer ≤ n

22. **11-12 C1 Recursive Functions**

Find $f(f(f(f(0))))$ given:

$$f(x)=\begin{cases}f(x+2)+1 & \textit{if } x<7\\ \text{x}-5 & \textit{if } x\geq 7\end{cases}$$

23. **12-13 C1 Recursive Functions**

Find $f(16)$ given:

$$f(x)=\begin{cases} f(f(x-2))+3 & if\ x>10 \\ f(x+1)\text{ - }1 & if\ x=10 \\ x\text{ - }4 & if\ x<10 \end{cases}$$

24. **12-13 C1 Recursive Functions**

Find $f(10,3)$ given:

$$f(x,y)=\begin{cases} f(x-2,y+1)+x & if\ x>5 \\ f(x-3,y+2)+\ \text{y} & if\ 0\le x\le 5 \\ x^2+x & if\ x<0 \end{cases}$$

25. **13-14 C1 Recursive Functions**

Find $f(12)$ given:

$$f(x)=\begin{cases} 2*f(x-3)-3 & if\ x>6 \\ f(x+2)+1 & if\ 4<x\le 6 \\ x+4 & if\ x\le 4 \end{cases}$$

26. **13-14 C1 Recursive Functions**

Find $f(10,2)$ given:

$$f(x,y)=\begin{cases} f(x-2,y+2)+2 & if\ x>y \\ f(x+1,y-1)\text{ - }1 & if\ x=\text{y} \\ xy & if\ x<y \end{cases}$$

27. **14-15 C1 Recursive Functions**

Find $f(14)$ given:

$$f(x)=\begin{cases} f(x-2)-1 & if\ x\ge 10 \\ f(x-1)+4 & if\ 6<x<10 \\ x-3 & if\ x\le 6 \end{cases}$$

28. **14-15 C1 Recursive Functions**

Find $f(f(f(10)))$ given:

$$f(x)=\begin{cases} f(x-3)\text{ - }1 & if\ x\ge 8 \\ f(x+1)+2 & if\ 5<x<8 \\ x+2 & if\ x\le 5 \end{cases}$$

29. **15-16 C1 Recursive Functions**

Find $f(6)$ given:

$$f(x)=\begin{cases}2*f(x-1)+f(x-2) & if\ x\geq 3\\ 0 & if\ x=2\\ 2 & if\ x=1\end{cases}$$

30. **15-16 C1 Recursive Functions**

Find $f(f(f(16)))$ given:

$$f(x)=\begin{cases}f(x-4)+2 & if\ x\geq 10\\ f(x+1)-1 & if\ 6<x<10\\ x+3 & if\ x\leq 6\end{cases}$$